

JEE MAIN REVISION MASTER: CHEMISTRY IN EVERYDAY LIFE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Chemistry In Everyday Life — Core Principles & Governing Laws	9 PYQs	Core Principle
Standard	Chemistry In Everyday Life — Applied Cases, Reactions & Traps	2 PYQs	Applied Topics
Standard	Chemistry In Everyday Life — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Chemistry In Everyday Life — Applied Cases, Reactions & Traps	$\text{COCl}_2 + 2\text{CHCl}_3 + \text{O}_2 \xrightarrow{\text{Sunlight}} 2\text{COCl}_2 + 2\text{HCl}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemistry In Everyday Life — Core Principles & Governing Laws (High Yield — 9 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Chemistry In Everyday Life.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Chemistry In Everyday Life PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the compounds of List-I with their effects in List-II: List-I List-II (a) Chloramphenicol (i) Malaria (b) Thyroxine (ii) Anaesthetic (...)" [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Which one of the following is a cationic detergent?..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Which one of the following gases converts haemoglobin into carboxy haemoglobin?..." [Ans: N/A]

3.2 Chemistry In Everyday Life — Applied Cases, Reactions & Traps (Standard Yield — 2 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "The compound is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Prolonged exposure of chloroform in humans may cause damage to liver. It is due to the formation of the following compound...." [Ans: N/A]

3.3 Chemistry In Everyday Life — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match the compounds of List-I with their effects in List-II: List-I List-II (a) Chloramphenicol (i) Malaria (b) Thyroxine (ii) Anaesthetic (...)" [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The compound is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match the compounds of List-I with their effects in List-II: List-I List-II (a) Chloramphenicol (i) Malaria (b) Thyroxine (ii) Anaesthetic (c) Chloroquine (iii) Goiter (d) Chloroform (iv) Typhoid fever

- (A) a-i, b-ii, c-iii, d-iv
- (B) a-iv, b-iii, c-i, d-ii
- (C) a-i, b-iii, c-iv, d-ii
- (D) a-iv, b-iii, c-ii, d-i

Correct Answer: None

Solution Analysis: Chloramphenicol is an antibiotic. It is used in the treatment of typhoid fever. Thyroxine is a thyroid hormone. Its deficiency is related to goiter, so it matches with goiter. Chloroquine is an antimalarial drug, so it is used for malaria. Chloroform has been used as an anaesthetic. So the correct matching is: (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii) Hence, the correct option is: Option B

Benchmark Problem 2: KCET 2024 (Q#2)

The compound is

- (A) sucralose
- (B) aspartame
- (C) saccharin
- (D) alitame

Correct Answer: None

Solution Analysis: The given structure is a phthalimide but no option of phthalimide is given. However, among the given options it is similar to saccharin. The structure of saccharin is